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WILDFIRE-S1S2-CANADA: A LARGE-SCALE SENTINEL-1/2 WILDFIRE BURNED AREA MAPPING DATASET BASED ON THE 2017-2019 WILDFIRES IN CANADA

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ABSTRACT

Wildfires vary across space and time, precisely and timely mapping on the wildfire affected areas is critical for wildfire management, population and property protection, and environmental impact assessment. In this study, we established a large-scale annotated wildfire burned area dataset based on freely available Sentinel-1 SAR and Sentinel-2 multispectral instrument (MSI) data and Canada Wildfire Burned Area Database. This dataset includes bi-temporal Sentinel-1 and Sentinel-2 images, which allows users to exploit remotely sensed data acquired in both optical and microwave domains. On the proposed dataset, we achieved the highest IoU score of 0.86 on the Sentinel-2 data with Siamese U-Net, and the highest IoU score of 0.80 on the Sentinel-1 data using U-Net with early fusion. The combined use of Sentinel-1 and Sentinel-2 failed to bring significant improvement compared to Sentinel-2 based results, but this dataset may have the potential to boost Sentinel-1 based results with Sentinel-2 data for near real-time wildfire progression mapping.

Index Terms— Wildfire Dataset, Change Detection, Burned Area Mapping, Deep Learning, U-Net, Siamese Network, Multi-source Data, Sentinel-1, Sentinel-2

1. INTRODUCTION

Wildfires are growing in both intensity and frequency, and are expanding their reaches around the globe, because of the global temperature rising caused by climate change that makes forest and fuels drier. In turn, the increasing wildfires may significantly impact local or even global climate considering the huge amounts of carbon dioxide released by them [1]. To reduce the economic loss and impacts on the environment from such destructive wildfires, it is critical to develop approaches for near real-time wildfire monitoring in a cost-effective manner.

Remotely sensed data from satellites have been widely used in monitoring changes happening on the earth surface and assessing their environmental impacts. The free access to Sentinel-1/2 satellite data makes a huge impact on the development of research and applications, and deep learning has achieved tremendous successes in remote sensing area [2]. However, deep learning relies heavily on the large-scale annotated dataset, but is still scarce for wildfire burned area mapping. Zhang *et. al* [3] investigated the potential of combined use of Sentinel-1 backscatter/coherence and Sentinel-2 surface reflectance data using siamese self-attention classifier based on 12 California wildfire events, which indicates that InSAR coherence contributes more significantly than SAR backscatter in burned area mapping. Rashkovetsky *et. al* [4] constructed a multisensor satellite wildfire dataset at the spatial resolution of 118m, which includes monotemoral Sentinel-1, Sentinel-2, SLSTR, and MODIS data, and compared the combination of different source of satellite data for burned area mapping.

In this study, we proposed a large-scale wildfire burned area dataset of 20m spatial resolution based on the 2017-2019 historical wildfires in Canada. The dataset provides bi-temporal images for both Sentinel-1 and Sentinel-2. We evaluated this dataset with deep learning-based semantic segmentation models such as U-Net and its Siamese variants [5].

2. DATA PREPARATION

2.1. Data Sources

The 2017-2019 wildfire events in Canada were based on the National Burned Area Database available in Natural Resources Canada. provided by the Canadian Wildland Fire Information System (CWFIS). The burned area polygons were generated mainly based on Landsat data, and it also includes both when and where wildfires started and ended, which are quite valuable in wildfire monitoring for both remote sensing and machine learning communities. To meet the needs, we exploited freely available Sentinel-1 C-Band SAR and Sentinel-2 multispectral satellite image data to establish a large-scale wildfire burned area dataset at the medium spatial resolution of 20m.

2.2. Workflow

Fig. 1 shows the workflow of data preparation on the S1S2-Wildfire dataset based on Google Earth Engine (GEE). Wildfire perimeter databases were used to derive the region of

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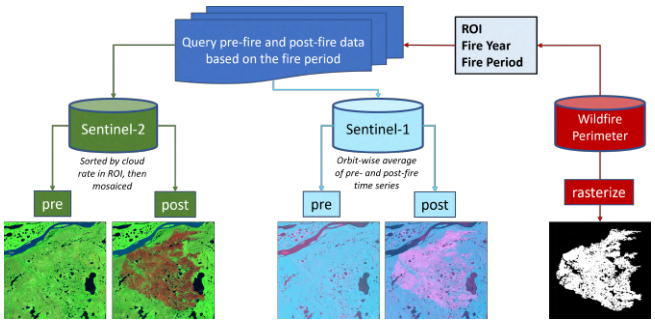

Fig. 1. Workflow of data preparation: pre-/post-year imagery of Sentinel-1/2 data and reference mask.

interests (ROI), fire year, and fire period. The Sentinel-1 time series data acquired over the same period (usually in summer) in one year before the fire year were averaged as the pre-fire image, while the post-fire images were generated by averaging the time series data acquired on the same orbits in the 2 month period after fire was contained. For Sentinel-2, to get cloud-free images, both the pre-fire and post-fire time series images were sorted by the property "CLOUDY_PIXEL_PERCENTAGE" in the descending order, and then mosaiced together as pre-fire image and post-fire image respectively. If the generated images are not in good quality visually, the post-fire period will be replaced by the images acquired in the same fire period one year after fire year, and the post-fire SAR and MSI images were ensured to be obtained from the same period. Additionally, wildfire perimeters were rasterized as the ground reference mask.

2.3. Data Export, Cleaning, and Tiling

After the preparation of the Sentinel-1/2 pre and post images for every wildfire event, we export them into Google Cloud Bucket together with rasterized ground reference masks at a spatial resolution of 20m in the format of COG. For each wildfire event, we exported both Sentinel-1 VH and VV polarized images acquired on different orbits and Normalized Difference Backscatter Index (NDBI, NDBI $= 10\log_{10}\frac{VV-VH}{VV+VH}$, where VH and VV are in power units) , and we exported 6 bands for Sentinel-2, including Red, Green, Blue, Near-Infrared (NIR), Short-Wave Infrared 1-/2 (SWIR-1 and SWIR-2), the most sensitive bands.

After downloading the exported data from Google Cloud Storage to a local machine, a visual inspection was conducted to exclude samples in poor quality: Each group of S1 pre-post, S2 pre-post images and the corresponding reference masks was converted to a visually perceivable format, as shown in Fig. 1. If either of these four images contains very large no-data areas, large clouds, or mosaicing caused artefacts, this group of images would be discarded. The remaining images were mirror padded into integer times of 256 both in height and width, and then tiled into patches of the shape 256 $\times$ 256 in a stride of 256 pixels, therefore adjacent patches have no overlap pixels except for the patches containing the pixels generated by mirror padding.

3. SENTINEL-1/2 WILDFIRE DATASET

After data cleaning, there are 328 wildfire events left, in which there are 132 events in 2017, 138 events in 2018, and 58 events for 2019. We used the 2017 and 2018 wildfire events as training set, and the 2019 wildfire events as testing set. For a single channel, there are 519,944,300 valid pixels in the training set, and 150,090,566 valid pixels in the testing set. By tiling all images and the reference masks into the shape of 256 $\times$ 256, the final wildfire burned area dataset contains 9052 groups of training patches and 2856 groups of testing patches. Each group of image patches includes the Sentinel-1 pre and post SAR images, Sentinel-2 pre and post MSI images, and the ground reference masks in the format of multi-channel Cloud-Optimised GeoTiff (COG), and the total size of this dataset is 60GiB. Fig. 1 shows some groups of images on five wildfire events, providing a visual perception on this wildfire burned area dataset.

3.1. Spatial and Temporal Coverage

As shown in Fig. 2, currently the wildfire burned area dataset was generated based on the Canada 2017-2019 wildfire events that have a burned area larger than 2000 hectares. In the near future, we will extend this dataset in both spatial and temporal dimensions, and the other data sources will be included too.

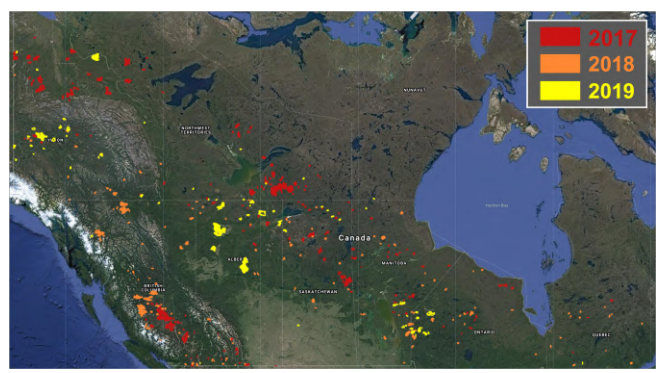

Fig. 2. Spatial and temporal coverage of the established wildfire burned area dataset in Canada.

Fig. 3 shows the land cover distribution of the proposed data in training and testing sets: The dominated land cover type is the closed forest with evergreen needle leaf in both training and testing sets, while open forest (class values: 121, 126) occupies about 18.5% in training set and 16.1% in testing set. 8.2% of land cover are shrubs in training set and 6.9% in testing set, and the rest main land covers include permanent

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water bodies, herbaceous vegetation, and closed forest with deciduous broadleaf.

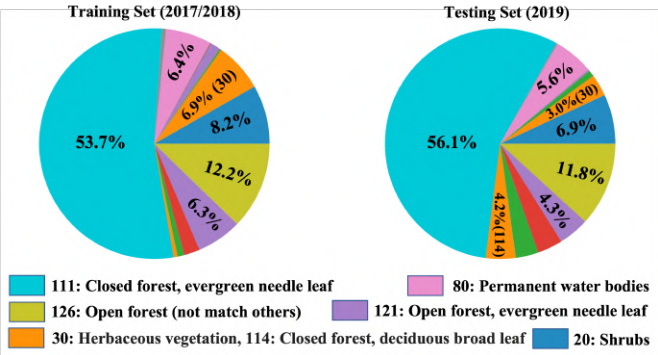

Fig. 3. Land cover distribution of the proposed wildfire burned area dataset (for the detailed description on the land cover, please refer to Copernicus Global Land Cover Layers).

3.2. Dataset Structure

This wildfire burned area dataset was organised in a tree-like structure with two main branches, i.e., *train, test*. In both the train and test folder, there are three subfolders, such as *S1, S2, and mask*, which denote Sentinel-1, Sentinel-2, and reference mask respectively. Both S1 and S2 folders contain two subfolders named 'pre' and 'post' that include pre-fire and post-fire image patches. The individual events can be identified by the filename of COG, i.e., event_id, and the tiled image patches were named by the following scheme *event_id_row_col.tif*, where *row* and *col* represent the location of this patch in the 256 $\times$ 256 grid of the original imagery before tiling.

4. APPLICATION TO BURNED AREA MAPPING

4.1. Semantic Change Detection Models

1) UNet-EF [5]: U-Net model with early fusion of input data, which means all input data are concatenated together along the channel dimension.
2) UNet-Siam-conc [5]: Siamese U-Net with two input streams (encoders), and the encoded features are concatenated together with decoded features along channel dimension. For single-sensor input, bi-temporal data are fed into two encoders separately, and these two streams may share the same encoders. For multi-sensor input, each encoder takes the stacked bi-temporal images of the same sensor as input.
3) UNet-Siam-diff [5]: The only difference between UNet-Siam-diff and UNet-Siam-conc is that the differenced features from two encoders are concatenated together with the decoder features at the same levels.

4.2. Experimental Results

Dice loss was taken as the default setting for model training, and we set the learning rate using the cosine annealing scheduler with the starting learning rate of 1e-4 and 2 warm-up steps. All models were trained within 100 epochs, the batch size of 16, and the weight decay of 0.01. The encoder architecture of UNet-EF is $2C$-16-16-MP-32-32-MP-64-64-MP-128-128-MP, where C denotes the number of channels of monotemporal image, and MP denotes MaxPool2d, based on which the decoder architecture can be derived. No data augmentation and dropout were applied, and all experiments were implemented with PyTorch and run on a single NVIDIA GTX3080 GPU with 10GiB memory.

Table 1. Quantitative Comparison on the Testing Set (SE denotes that Siamese architecture shares the same encoders).

Sat.	Method	SE	F1	IoU
S2	UNet-EF-6bands	-	0.9167	0.8462
	UNet-EF	-	0.9145	0.8425
	UNet-Siam-conc	Yes	**0.9258**	**0.8618**
	UNet-Siam-conc	No	0.9114	0.8373
	UNet-Siam-diff-abs	Yes	**0.9274**	**0.8646**
	UNet-Siam-diff	Yes	**0.9250**	**0.8604**
	UNet-Siam-diff	No	0.9153	0.8438
S1	UNet-EF	-	**0.8866**	**0.7963**
	UNet-EF-VH	-	**0.8508**	**0.7403**
	UNet-EF-VV	-	0.8245	0.7014
	UNet-EF-NDBI	-	0.7719	0.6285
	UNet-Siam-conc	Yes	0.8533	0.7441
	UNet-Siam-conc	No	0.8512	0.7410
	UNet-Siam-diff-abs	Yes	**0.8784**	**0.7832**
	UNet-Siam-diff	Yes	0.8720	0.7731
	UNet-Siam-diff	No	0.8390	0.7227
S1+S2	UNet-EF	-	0.9191	0.8503
	UNet-Siam-conc	No	**0.9227**	**0.8565**

Table 1 summarises the quantitative comparison of different configurations on the testing sets, without specification, S2 models take B4, B8, and B12 as the default input while S1 models take VH, VV, and NDBI. On the S2 data, Siamese U-Net achieved significantly higher F1 and IoU scores with weight-sharing encoders (SE). Without weight sharing, UNet-Siam-diff obtained a similar accuracy to UNet-EF. On the S1 data, UNet-Siam-diff with SE performs better than UNet-Siam-conc, and UNet-EF achieved a slightly higher IoU score than UNet-Siam-diff-abs (absolute feature difference). Based on UNet-EF, VV performs the best among VH, VV, and NDBI. Based on S1 and S2, UNet-EF didn't improve significantly compared to S2-based results, and UNet-Siam-conc

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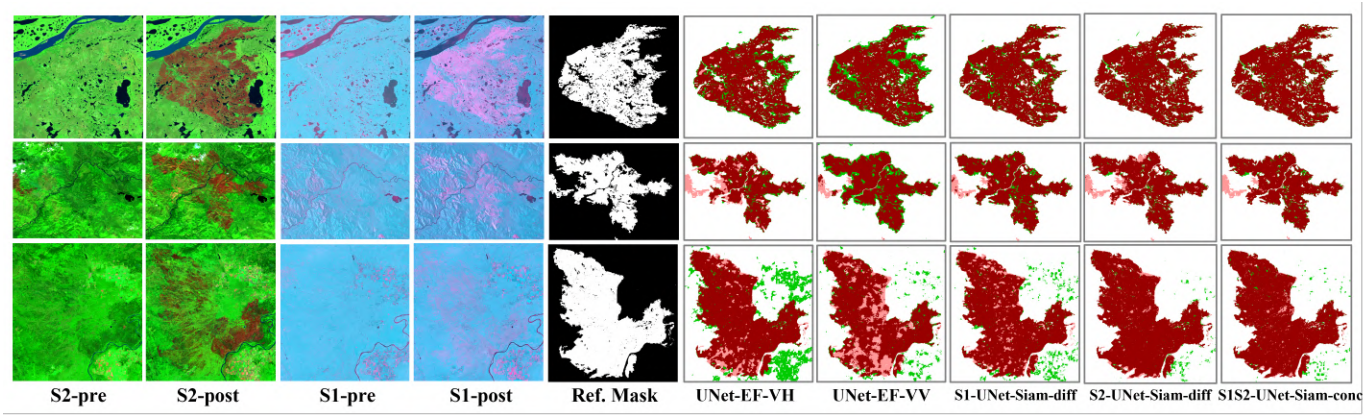

Fig. 4. Visual comparison of three wildfire events (true positive in dark red, false positive in green, and false negative in pink).

achieved similar results to UNet-EF without weight sharing. Fig. 4 shows pre-post images, reference masks, and visual results on three testing wildfire events.

4.3. Discussion

Table 1 indicates that UNet-Siam-diff architecture and UNet-EF are two suitable methods for wildfire burned area mapping, which is consistent with the findings on three other change detection dataset in [5]. Rashkovetsky *et. al* [4] achieved the IoU score of 0.71 and 0.39 based on monotemporal S2 and S1 post-fire images, our results demonstrated that bi-temporal SAR or MSI images are able to significantly boost IoU score to 0.86 and 0.80 for S2 and S1 respectively. By fusing monotemporal or time series S1 backscatter with S2 multispectral images acquired in clear conditions, no improvement was observed compared to S2-based results [3, 4]. The same conclusion was drawn on the proposed bi-temporal Sentinel-1/2 dataset, since cloud-free Sentinel-2 data itself could provide rich spectral information to achieve satisfying results in burned area mapping.

5. CONCLUSION

In this paper, we proposed a large-scale annotated wildfire burned area dataset which includes bi-temporal Sentinel-1 SAR and Sentinel-2 MSI data and rasterized wildfire perimeters. We evaluated this dataset with three deep learning architectures, i.e., UNet-EF, UNet-Siam-conc, and UNet-Siam-diff, and found that UNet-EF achieved the highest IoU score on Sentinel-1 data, while UNet-Siam-diff achieved the highest IoU score on the Sentinel-2 data with weight-sharing encoders. Based on UNet-EF, we found VH performed best, followed by VV, and combining VH, VV, and NDBI can significantly boost IoU score from 0.62-0.74 to 0.80. Though the combined use of Sentinel-1 and Sentinel-2 failed to improve the IoU score compared to Sentinel-2 based results in burned area mapping, the proposed wildfire burned area dataset may provide new opportunities to exploit Sentinel-2 data to boost Sentinel-1 SAR based results for near real-time wildfire progression mapping.

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